When Does a Scale Factor Get Squared or Cubed? Answer Key

High School Geometry

Quick Answers

1. (a) correct enlarged area = 216 cm^2, —
2. 8000 cm^3
3. (a) correct volume of the larger cone = 1406.25 cm^3, —
4. 2.5
5. (a) linear scale factor = 1.67, (b) surface area of the larger solid = 175 cm^2
6. (a) real capacity = 12000 L, (b) real sheet-metal area = 320 m^2, —

What is verified

8 of 11 answers machine-checked against SymPy, across 6 problems.
— marks an answer only you can judge — problems 1, 3, 6. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry
HSG-GMD.A.3 – *problems 1, 2, 3, 4, 5*
HSG-MG.A.2 – *problem 6*
Difficulty 2–5 of 5 across 11 tagged checks.

Common wrong answers

1. *If they got 72: multiplied the area by the linear scale factor instead of by its square*
2. *If they got 500: scaled the volume by the linear factor instead of by its cube*
3. *If they got 225: multiplied the volume by the ratio of the heights instead of by its cube*
4. *If they got 6.25: used the surface-area ratio itself as the linear scale factor*

1. Rectangle of area 24 cm^2, every length multiplied by 3. Dev writes $24 \times 3 = 72$.

Solution (a): Enlarging every length by 3 stretches the rectangle in *two* directions at once: the length becomes 3 times longer and the width becomes 3 times longer, so the area picks up a factor of 3 twice.

$$A' = 24 \times 3^2 = 24 \times 9$$

So $\boxed{A' = 216\text{ cm}^2}$

Solution (b) — instructor-judged. A model response: Dev used the linear scale factor where the *squared* scale factor belongs. Because both dimensions are stretched by 3, the area is multiplied by $3 \times 3 = 9$, not by 3. His answer of 72 cm^2 is the area of a rectangle only three times as big, which cannot be right for a shape whose sides have all tripled.

Full credit: names k^2 as the area factor and ties it to two dimensions being stretched. *Half credit:* states the rule without the reason. *No credit:* redoes the arithmetic with no account of the error.

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2. Cube of edge 5 cm, every length multiplied by 4. Find the enlarged volume.

Solution: The original volume is $5^3 = 125\ \text{cm}^3$. A volume is stretched in three directions, so it is multiplied by $4^3 = 64$.

$$V' = 125 \times 64$$

So $\boxed{V' = 8000\ \text{cm}^3}$

Check the other way: the new edge is $5 \times 4 = 20$ cm, and $20^3 = 8000$ ✓

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3. Similar cones, heights 6 cm and 15 cm; smaller volume 90 cm^3. Mia writes $90 \times \frac{15}{6} = 225$.

Solution (a): The linear scale factor is $\frac{15}{6} = 2.5$. Volumes of similar solids scale by the *cube* of that factor.

$$V' = 90 \times \left(\frac{15}{6}\right)^3 = 90 \times 15.625$$

So $\boxed{V' = 1406.25\ \text{cm}^3}$

Solution (b) — instructor-judged. A model response: Mia multiplied by the linear scale factor 2.5 instead of by $2.5^3 = 15.625$. Similar solids have every length scaled by 2.5, so a volume — which is a product of three lengths — is scaled by 2.5 three times over. Her answer is too small by a factor of $2.5^2 = 6.25$.

Full credit: identifies the linear factor as 2.5 and states the cube rule. *Half credit:* names the cube rule without computing the linear factor. *No credit:* says the answer is wrong with no rule.

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4. Similar prisms with surface areas 48 cm^2 and 300 cm^2. Find the linear scale factor.

Solution: Surface areas scale by k^2, so the ratio of areas is k^2, and k is its square root — not the ratio itself.

$$k^2 = \frac{300}{48} = 6.25 \qquad k = \sqrt{6.25}$$

So $\boxed{k = 2.5}$

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5. Similar solids of volume 54 cm^3 and 250 cm^3; smaller surface area 63 cm^2.

Solution (a): Volumes scale by k^3, so take the cube root of the volume ratio:

$$k = \sqrt[3]{\frac{250}{54}} = \sqrt[3]{\frac{125}{27}} = \frac{5}{3} = 1.6666\ldots$$

So $\boxed{k \approx 1.67}$

Solution (b): Surface areas scale by k^2, and here $k = \frac{5}{3}$ exactly, so

$$S' = 63 \times \left(\frac{5}{3}\right)^2 = 63 \times \frac{25}{9}$$

giving $\boxed{S' = 175\ \text{cm}^2}$

Note that the exact fraction $\frac{5}{3}$ was used here rather than the rounded 1.67 — rounding early would have given 175.6.

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6. Model tank at 1 : 20. (a) Model holds 1.5 L. (b) Model shell uses 0.8 m^2. (c) Why different multipliers?

Solution (a): A capacity is a volume, so it scales by $20^3 = 8000$.

$$1.5 \times 8000$$

So $\boxed{12000\ \text{L}}$

Solution (b): A sheet-metal area scales by $20^2 = 400$.

$$0.8 \times 400$$

So $\boxed{320\ \text{m}^2}$

Solution (c) — instructor-judged. A model response: the single scale factor 20 acts on every *length*. A surface is two-dimensional, so it is stretched twice over and grows by $20^2 = 400$. A capacity is three-dimensional, so it is stretched three times over and grows by $20^3 = 8000$. One linear ratio therefore produces two different multipliers, and they differ by exactly one more factor of 20.

Full credit: explains the difference through the number of dimensions and gives both multipliers. *Half credit:* gives both multipliers with no reason. *No credit:* says only that volume grows more than area.